



EduSync

Online Learning Management Platform

Software Requirements Specification

Functional Requirements, User Stories & Acceptance Criteria

Version 2.1

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PREPARED BY Product & Engineering Team	ORGANISATION EduSync Inc.
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20

USER STORIES

8

FEATURE AREAS

3

USER ROLES

60+

FORM FIELDS

FEATURE AREAS COVERED



Authentication & Accounts



Course Discovery & Enrollment



Assignments & Submissions



Quizzes & Assessments



Course Management



Learner Profile & Progress



Notifications & Messaging



Administration

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Introduction & Overview

1.1 PURPOSE

This document describes the functional requirements for the **EduSync Online Learning Management Platform**. It is intended for developers, QA engineers, UX designers, and stakeholders to understand the expected system behaviour and all form interactions.

Each user story is accompanied by detailed acceptance criteria and a list of required form fields with their validation constraints.

1.2 PROJECT SCOPE

EduSync is a full-featured web-based LMS supporting learners, instructors, and administrators. Key capabilities include:

- Course enrollment & discovery
- Live and on-demand sessions
- Assignment submission & grading
- Timed quizzes & assessments
- Certification & badge management
- Platform administration & billing

2.1 SYSTEM OVERVIEW

The EduSync platform connects learners and instructors through a unified portal. Learners can discover courses, enrol, complete assignments, take quizzes, and earn certificates. Instructors can create and manage courses, track student progress, and grade submissions. Administrators oversee user management, billing, and platform-wide settings.

2.2 USER ROLES

● Learner

A registered student who browses, enrolls in, and completes courses. Can submit assignments, take quizzes, and earn certificates.

● Instructor

A content creator who publishes and manages courses, creates quizzes, grades submissions, and communicates with learners.

● Administrator

A platform manager with full access to all user data, billing, subscription plans, and platform-wide settings and announcements.

Authentication & Account Management



Authentication & Account Management

4 User Stories

Registration, login, password reset, and instructor onboarding flows.

US-001

Learner Registration

As a **new learner**, I want to **create an account with my personal details and email**, so that I can access courses and track my learning progress.

ACCEPTANCE CRITERIA

- ✓ Form collects: `full_name`, `email`, `password`, `confirm_password`, `date_of_birth`, `country`
- ✓ Email must match standard format `user@example.com`
- ✓ Password ≥ 8 characters, must include one uppercase letter, one number, and one special character
- ✓ Confirm password must match the password field exactly
- ✓ Date of birth must be valid and learner must be at least 13 years old
- ✓ Country selected from a predefined dropdown list
- ✓ On successful submission, a verification email is sent to the provided address

`full_name` text

`email` email

`password` password

`confirm_password` password

`date_of_birth` date

`country` select

US-002

Learner Login

As a **registered learner**, I want to **sign in using my email and password**, so that I can access my enrolled courses and dashboard.

ACCEPTANCE CRITERIA

- ✓ Form collects: `email`, `password`
- ✓ Optional `remember_me` checkbox to persist the session for 30 days
- ✓ Clicking "Forgot Password" redirects to the password reset flow
- ✓ After 5 consecutive failed attempts, show a generic error and lock the account for 15 minutes

`email` email

`password` password

`remember_me` checkbox

US-003

Password Reset

As a **learner who has forgotten their password**, I want to **reset it using my email**, so that I can regain access to my account.

ACCEPTANCE CRITERIA

- ✓ Forgot-password form collects: `registered_email`
- ✓ A reset link is sent to the email address within 2 minutes
- ✓ New password form collects: `new_password` and `confirm_new_password`
- ✓ New password must meet the same complexity rules as the registration password
- ✓ Reset links expire after 1 hour

`registered_email` email

`new_password` password

`confirm_new_password` password

US-004

Instructor Registration

As a **subject-matter expert**, I want to **register as an instructor with my professional details**, so that I can create and publish courses on the platform.

ACCEPTANCE CRITERIA

- ✓ Form collects: `full_name`, `email`, `password`, `professional_title`, `area_of_expertise`, `years_of_experience`, `biography`, `linkedin_url`, `profile_photo`
- ✓ Area of expertise: one of Technology, Business, Design, Marketing, Science, Language, Arts, Health
- ✓ Years of experience must be a positive integer between `0` and `50`
- ✓ LinkedIn URL must start with `https://www.linkedin.com/in/`
- ✓ Biography must not exceed 500 characters
- ✓ Profile photo must be JPEG or PNG, maximum 2 MB

`full_name` text

`email` email

`password` password

`professional_title` text

`area_of_expertise` select

`years_of_experience` number

`biography` textarea

`linkedin_url` url

`profile_photo` file

Course Discovery & Enrollment



Course Discovery & Enrollment

Search, filter, and enrol in paid and free courses.

3 User Stories

US-005

Course Search

As a **learner**, I want to **search for courses using keywords and filters**, so that I can quickly find courses relevant to my learning goals.

ACCEPTANCE CRITERIA

- Form collects: keyword , category , difficulty_level , price_min , price_max , rating , language
- Keyword field requires a minimum of 2 characters
- Difficulty level: Beginner / Intermediate / Advanced
- Price range minimum must not exceed price range maximum
- Rating filter: 1-5 stars
- Results must update within 1 second of applying filters
- Supports sorting by: Relevance, Newest, Highest Rated, Most Popular

keyword text ≥2

category select

difficulty_level radio

price_min number

price_max number

rating number 1-5

language select

US-006

Course Enrollment (Paid)

As a **learner**, I want to **enrol in a course by completing the payment process**, so that I can start learning immediately after successful payment.

ACCEPTANCE CRITERIA

- Form collects: course_id (hidden), cardholder_name , card_number , expiry_month , expiry_year , cvv , billing_postal_code
- Card number must be 16 digits and pass Luhn algorithm validation
- Expiry date must be in the future
- CVV must be 3 digits for Visa/Mastercard, 4 digits for Amex
- On successful payment, learner receives a confirmation email with receipt
- Learner is immediately added to the course roster

course_id hidden

cardholder_name text

card_number text 16d

expiry_month select

expiry_year select

cvv text 3-4d

billing_postal_code text

US-007

Free Course Enrollment

As a **learner**, I want to **enrol in a free course with a single click**, so that I can access free content without payment details.

ACCEPTANCE CRITERIA

- ✓ Confirmation form collects: `consent_checkbox` ("I agree to the Terms of Service and Privacy Policy")
- ✓ Consent checkbox is required before enrollment can proceed
- ✓ Upon enrollment, learner is redirected to the course overview page

`consent_checkbox` checkbox req.

Assignment & Submission



Assignment & Submission

2 User Stories

Learner submission and instructor grading workflows.

US-008

Assignment Submission

As an **enrolled learner**, I want to **submit my assignment before the deadline**, so that my instructor can grade my work and provide feedback.

ACCEPTANCE CRITERIA

- ✓ Form collects: `assignment_id` (hidden), `submission_title`, `submission_notes`, `file_attachment`, `submission_date` (auto-populated), `integrity_checkbox`
- ✓ Submission title is required, between 5 and 150 characters
- ✓ Submission notes are optional, up to 1,000 characters
- ✓ File attachment required (PDF, DOC, DOCX, or ZIP), maximum 50 MB
- ✓ Integrity checkbox ("I confirm this is my own work") is required
- ✓ Submissions after the deadline are flagged as "Late"

`assignment_id` hidden

`submission_title` text 5-150

`submission_notes` textarea ≤1000

`file_attachment` file ≤50MB

`submission_date` date auto

`integrity_checkbox` checkbox req.

US-009

Assignment Grading

As an **instructor**, I want to **grade a learner's submission and provide written feedback**, so that the learner can understand their performance.

ACCEPTANCE CRITERIA

- ✓ Form collects: `submission_id` (hidden), `grade`, `feedback_text`, `grading_date` (auto-populated)
- ✓ Grade must be a whole number between 0 and 100
- ✓ Feedback text is required, between 10 and 2,000 characters, before a grade can be saved
- ✓ A grade below 50 is automatically flagged as "Failing"
- ✓ Learner receives an email notification when grading is complete

`submission_id` hidden

`grade` integer 0-100

`feedback_text` textarea 10-2000

`grading_date` date auto

Quizzes & Assessments



Quizzes & Assessments

2 User Stories

Timed quiz attempts and instructor quiz creation.

US-010

Quiz Attempt

As an **enrolled learner**, I want to **take a timed multiple-choice quiz**, so that I can test my understanding of the course material.

ACCEPTANCE CRITERIA

- ☒ Form collects: `quiz_id` (hidden), `learner_id` (hidden), `selected_answers[]` (radio per question), `time_taken` (auto-captured)
- ☒ Learner has a maximum of 3 attempts per quiz
- ☒ A countdown timer starts when the quiz begins; duration set by instructor
- ☒ Auto-submit triggers when the timer reaches zero
- ☒ Results displayed immediately after submission with correct answers shown

`quiz_id` hidden`learner_id` hidden`selected_answers[]` radio req.`time_taken` integer auto

US-011

Quiz Creation

As an **instructor**, I want to **create a quiz with multiple-choice questions**, so that I can assess learner understanding after each module.

ACCEPTANCE CRITERIA

- ☒ Form collects: `quiz_title`, `course_module`, `time_limit`, `passing_score`, `max_attempts`, `show_correct_answers`
- ☒ Quiz title required, 5–100 characters
- ☒ Time limit in minutes: 5–180
- ☒ Passing score percentage: 1–100
- ☒ Maximum attempts: 1–5
- ☒ At least 3 questions must be added before publishing; each question needs exactly 4 choices with 1 correct

`quiz_title` text 5–100`course_module` select`time_limit` integer 5–180`passing_score` integer 1–100`max_attempts` integer 1–5`show_correct_answers` toggle

Course Management (Instructor)



Course Management

2 User Stories

Course creation, metadata, pricing, and module organisation.

US-012

Course Creation

As an **instructor**, I want to **create a new course with complete metadata and pricing**, so that learners can find and enrol in my course.

ACCEPTANCE CRITERIA

- ✓ Form collects: `course_title`, `subtitle`, `description`, `category`, `difficulty_level`, `language`, `price`, `promotional_price`, `thumbnail`, `preview_video_url`, `learning_outcomes[]`
- ✓ Course title: 5–120 characters, must be unique within the instructor's library
- ✓ Description minimum 200 characters; subtitle 10–200 characters
- ✓ Price: \$0.00–\$999.99 (0 = free); at most 2 decimal places
- ✓ Promotional price, if provided, must be less than the regular price
- ✓ Thumbnail: JPEG/PNG, max 2 MB; at least 3 learning outcome bullet points

course_title text 5–120

subtitle text 10–200

description textarea ≥200

category select

difficulty_level radio

language select

price decimal 0–999.99

promotional_price decimal opt.

thumbnail file ≤2MB

preview_video_url url

learning_outcomes[] array ≥3

US-013

Course Module Creation

As an **instructor**, I want to **organise my course content into named modules**, so that learners can follow a structured learning path.

ACCEPTANCE CRITERIA

- ✓ Form collects: `module_title`, `module_description`, `module_order`, `estimated_duration`
- ✓ Module title: 5–100 characters, unique within the same course
- ✓ Module description optional, up to 300 characters
- ✓ Estimated duration: 1–600 minutes
- ✓ Module order is auto-incremented but editable as a positive integer
- ✓ Changes are auto-saved every 30 seconds

module_title text 5–100

module_description textarea ≤300

module_order integer auto

estimated_duration integer 1–600

Learner Profile & Progress



Learner Profile & Progress

2 User Stories

Profile updates, preferences, and certificate requests.

US-014

Learner Profile Update

As a **learner**, I want to **update my profile with a photo, bio, and learning preferences**, so that instructors and peers can learn more about me.

ACCEPTANCE CRITERIA

- ✓ Form collects: `display_name`, `profile_photo`, `bio`, `preferred_topics[]`, `weekly_goal_hours`, `notification_email`, `notification_push`
- ✓ Display name: 2–50 characters; letters, spaces, hyphens, and apostrophes only
- ✓ Profile photo optional: JPEG/PNG, max 2 MB
- ✓ Bio optional, up to 300 characters
- ✓ Preferred learning topics: multi-select, up to 5 tags
- ✓ Weekly learning goal: whole number between 1 and 40 hours

`display_name` text 2–50`profile_photo` file opt.`bio` textarea ≤300`preferred_topics[]` multi-select ≤5`weekly_goal_hours` integer 1–40`notification_email` checkbox`notification_push` checkbox

US-015

Certificate Request

As a **learner who has completed a course with a passing grade**, I want to **request my completion certificate**, so that I can share my achievement on my portfolio or LinkedIn.

ACCEPTANCE CRITERIA

- ✓ Form collects: `legal_name`, `course_id` (hidden), `completion_date` (auto-populated), `delivery_format`, `name_change_reason` (conditional)
- ✓ Full legal name must match the learner's registered name, or a reason must be selected from: Name Change, Professional Name, Other
- ✓ Delivery format: PDF / Digital Badge / Both
- ✓ Certificate generated within 24 hours of request

`legal_name` text`course_id` hidden`completion_date` date auto`delivery_format` radio`name_change_reason` select cond.

Notifications & Communication



Notifications & Communication

2 User Stories

Instructor messaging and notification preference configuration.

US-016

Contact Instructor

As a **learner**, I want to **send a direct message to my course instructor**, so that I can ask questions about the course material.

ACCEPTANCE CRITERIA

- ✓ Form collects: `recipient_instructor_id` (hidden), `subject`, `message_body`, `file_attachment` (optional)
- ✓ Subject required, 5–100 characters
- ✓ Message body minimum 20 characters, maximum 2,000 characters
- ✓ Optional file attachment: PDF/JPG/PNG, max 5 MB
- ✓ Learner may not send more than 10 messages per day per instructor

`recipient_instructor_id` hidden`subject` text 5–100`message_body` textarea 20–2000`file_attachment` file opt. ≤5MB

US-017

Notification Preferences

As a **learner**, I want to **configure which notifications I receive and how**, so that I only get alerts that are relevant to me.

ACCEPTANCE CRITERIA

- ✓ Form collects: `email_notifications`, `push_notifications`, `new_announcements`, `assignment_reminders`, `quiz_availability`, `grade_posted`, `weekly_digest`, `preferred_notification_time`
- ✓ At least one notification channel (email or push) must be enabled
- ✓ Preferred notification time (HH:MM) is only active when email or push is enabled

`email_notifications` toggle`push_notifications` toggle`new_announcements` checkbox`assignment_reminders` checkbox`quiz_availability` checkbox`grade_posted` checkbox`weekly_digest` checkbox`preferred_notification_time` time HH:MM

Administration



Administration

3 User Stories

User management, announcements, and subscription plan administration.

US-018 User Account Management

As a **platform administrator**, I want to **search and manage user accounts**, so that I can resolve issues and enforce platform policies.

ACCEPTANCE CRITERIA

- ✓ Form collects: `search_query` , `role_filter` , `status_filter` , `date_from` , `date_to` , `results_per_page`
- ✓ Role filter: All / Learner / Instructor / Admin
- ✓ Status filter: Active / Suspended / Pending Verification
- ✓ "From" date must not be after "to" date
- ✓ Results per page: 10 / 25 / 50 / 100; must load within 2 seconds

`search_query` text `role_filter` select `status_filter` select `date_from` date `date_to` date `results_per_page` select

US-019 Platform Announcement

As a **platform administrator**, I want to **publish a site-wide announcement to all users**, so that I can inform them of important updates or maintenance.

ACCEPTANCE CRITERIA

- ✓ Form collects: `announcement_title` , `announcement_body` , `target_audience` , `publish_at` , `expires_at` , `priority_level`
- ✓ Title: 5–100 characters; body: 20–1,000 characters
- ✓ Target audience: All Users / Learners Only / Instructors Only
- ✓ Expiry date must be after publish date
- ✓ Priority level: Info / Warning / Critical — Critical triggers immediate push notification

`announcement_title` text 5–100 `announcement_body` textarea 20–1000 `target_audience` select `publish_at` datetime `expires_at` datetime `priority_level` radio

Subscription Plan Management

As a **platform administrator**, I want to **create and edit subscription plans**, so that learners have clear pricing options for premium access.

ACCEPTANCE CRITERIA

- ✓ Form collects: `plan_name` , `billing_cycle` , `price_per_cycle` , `max_courses_per_month` , `includes_live_sessions` , `trial_period_days` , `plan_description`
- ✓ Plan name: 3-50 characters, unique across all active plans
- ✓ Billing cycle: Monthly / Quarterly / Annual
- ✓ Price: \$0.01-\$9,999.99 (greater than zero unless free tier)
- ✓ Trial period: 0-30 days (0 = no trial); plan description up to 500 characters

`plan_name` text 3-50

`billing_cycle` select

`price_per_cycle` decimal

`max_courses_per_month` integer

`includes_live_sessions` toggle

`trial_period_days` integer 0-30

`plan_description` textarea ≤500

Non-Functional Requirements

Performance

- › All form submissions must receive a server response within 3 seconds.
- › Search results must appear within 1 second of applying filters.
- › The platform must support 10,000 concurrent users without performance degradation.

Security

- › All form data must be transmitted over HTTPS.
- › Passwords hashed using bcrypt with a minimum cost factor of 10.
- › File uploads scanned for malware before storage.
- › Session tokens expire after 24 hours of inactivity.

Accessibility

- › All form fields must have associated ARIA labels.
- › Forms must be fully navigable by keyboard alone.
- › Error messages must be screen-reader compatible.
- › Colour contrast must meet WCAG 2.1 AA standards.

Availability

- › Platform uptime SLA: 99.9% ($\leq$ 8.7 hours downtime per year).
- › Scheduled maintenance during off-peak hours (02:00–04:00 UTC).
- › All user data backed up daily with a 30-day retention window.

Appendix — Glossary & Related Documents

APPENDIX A — GLOSSARY

Acronym	Definition
LMS	Learning Management System — a software platform for administering, documenting, tracking, and delivering educational courses.
SRS	Software Requirements Specification — a document that describes what a software system should do.
WCAG	Web Content Accessibility Guidelines — international standards for making web content accessible to people with disabilities.
API	Application Programming Interface — a set of protocols for building and integrating software applications.
SLA	Service Level Agreement — a commitment between a service provider and a client defining the level of service expected.
JWT	JSON Web Token — a compact, URL-safe means of representing claims to be transferred between two parties.
CVV	Card Verification Value — a 3- or 4-digit security code printed on a payment card.
ARIA	Accessible Rich Internet Applications — a set of attributes that define ways to make web content more accessible.

APPENDIX B — RELATED DOCUMENTS

- EduSync API Design Specification v1.4**
Defines all REST API endpoints, request/response schemas, authentication headers, and error codes used by the frontend and third-party integrations.
- EduSync Database Schema Reference v2.0**
Entity-relationship diagram and table definitions for all Prisma models, including indexes, foreign key constraints, and soft-delete conventions.
- EduSync UI/UX Style Guide v3.1**
Design tokens, component library, typography scale, colour palette, spacing system, motion principles, and accessibility guidelines for the EduSync design system.

